Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/15/2008
PI Name:	Ullrich, Robert Ph.D.		
Project Title:	NSCOR: Radiation Leukemogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bullrich@utmb.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	409-747-1935
Organization Name:	University of Texas Medical Branch		
PI Address 1:	301 University Blvd		
PI Address 2:	Comprehensive Cancer Center, MS 1048		
PI Web Page:			
City:	Galveston	**State:**	TX
Zip Code:	77555-5302	**Congressional District:**	14
Comments:	NOTE: PI moved to UTMB from Colorado State University in late 2008 (6/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	NSCOR 03-OBPR-02
Start Date:	10/01/2003	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bailey, Susan (Colorado State University) Bedford, Joel (Colorado State University) Belmont, John (Baylor College of Medicine) Cai, Wei-Wen (Baylor College of Medicine) Ehrhart, E.J. (Colorado State University) Nagarajan, Lalitha (U.T., M.D. Anderson. Cancer Center) Story, Michael (University of Texas, M. D. Anderson Cancer Center) Weil, Michael (U.T., M.D. Anderson. Cancer Center)		
Grant/Contract No.:	NAG9-1569		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of this program is to provide information required to develop a rational scientific basis for estimation of risks for leukemogenesis in humans from exposure to radiation during space flight. This will be accomplished using several coordinated research activities. A key activity will be to develop quantitative data on the relative leukemogenic effectiveness of specific HZE particles and of high-energy protons compared with gamma rays in a mouse model of radiation-induced acute myeloid leukemia (AML). Extrapolation of risks from this mouse model to risks in humans requires additional information, and a major component of research activities in this program are designed to provide such information. A major research effort of this program is focused on studies of cellular and molecular mechanisms and processes involved in radiation-induced AML. These studies will not only improve the basic understanding of the pathogenesis of AML but will also provide information on cellular and molecular biomarkers that may be useful in predicting individuals at increased risk. These activities will focus on the use of molecular and molecular cytogenetic approaches, the development of new model systems, and comparative analyses of murine and human AML to facilitate the translation of results obtained in mice to human AMLs. Problems of extrapolation are not limited to biological factors. There are clear dosimetric differences that also must be considered when extrapolating data obtained from mouse studies to human risks. To address these issues this project also has an important physics and dosimetry component that includes modeling of dose to target cells in murine and human bone marrow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Not Available
Task Progress:	The four projects comprising this NSCOR are briefly described below. Project 1. Induction of AML in CBA/CaJ mice (project leader, Dr. Robert. L. Ullrich). This project is designed to quantitatively compare the leukemogenic effects of irradiation with gamma-rays, select HZE particles, and protons using the CBA/CaJ murine model of AML. These experiments are designed to provide sufficient quantitative data to permit the determination of their relative effectiveness by comparing slope constants obtained over specified dose ranges rather than fully defining the dose response relationships for each of these radiations. Project 2. Quantification and characterization of specific cytogenetic changes associated initiation, progression, and development of AML (project leader, Dr. Joel S. Bedford). RBEs for a particular radiation quality can depend very much on a number of factors such as the biological effect used for comparisons, as well as the level of effect and conditions such as chemical environment. Likewise, carcinogenesis in general depends on a number of factors including not only initiating events, but also a number of tissue and organ specific conditions. Potential modifying effects associated with tissue and organ specific effects may well differ between mouse and man. On this basis, quantification and characterization of specific cytogenetic changes that are known to be important early events, and understanding effects on persistence and expansion of cells containing such alterations should be an important bridge between the mouse and human AML. Experiments in Project 2 are designed to develop such quantitative information on early events and characterize progression as a function of radiation quality. Project 3. Molecular and cytogenetic fingerprints and biomarkers. (project leaders, Drs. Joel Bedford, Susan Bailey, Wei-Wen Cai, and Michael Story). The goals of Project 3 deal with investigations aimed at gaining a better understanding of factors at the cell and molecular level that may be involved in radiation-induced murine AML in general, and for HZE radiation exposures in particular. The overall long-term focus of the studies in Project 3 is on developing comparative information on radiation leukemogenesis in mice and humans. The studies outlined are subdivided into five sub-projects. Project 3A will investigate the possible role of the structural organization of chromatin in cytogenetic damage associated with AML. Project 3B will utilize microarray-based comparative genomic hybridization of normal vs. tumor DNA to identify regions of genomic deletions and amplifications common to the radiation-induced AML. Project 3C will investigate the possibility that HZE radiation induces chromosomal instabilities in the clonal progeny of surviving bone marrow cells after mice are exposed to HZE radiations. This sub-project also would investigate the long term stability of complex aberrations in such spleen colonies, since exposure to other high LET radiations suggest that even for low dose exposures, more than 90% of all aberrations are complex. Project 3D will investigate changes in gene expression by array technologies, in leukemogenic or pre-leukemogenic cells with a focus on changes in expression that may correlate or track with genomic changes seen in other studies in this program of studies. Project 4. Molecular and cytogenetic targets in murine and human AML. (Project leaders, Drs. Joel Bedford, Robert Ullrich, Lalitha Nagarajan, and John Belmont). The goals of Project 4 are to identify common cytogenetic and molecular targets in mice and humans that are involved in the development of radiation-induced AML. Studies in this project are closely integrated with studies in Project 3 that focus on cell and molecular factors of importance for murine leukemogenesis. Project 4 provides the link between murine and human AML that will be necessary to extrapolate from mouse studies to estimate human risks. Project 4 contains two sub-projects. Project 4A, deals with more complete cytogenetic characterization of ã-ray and HZE induced murine AML, beyond the known chromosome 2 deletion, and aims to compare mouse and human cytogenetic changes involved in leukemogenesis. The goal of this comparison is to identify common syntenic regions or conserved linkage segments present at the breakpoints of translocations or deleted regions of the two species. Project 4B, involves the development of a human/mouse hybrid model of human leukemogenesis and the use of this model to quantify AML induction by HZE and gamma-rays. This project will also determine whether there are cytogenetic changes in irradiated human hematopoietic cells that may be predictive of leukemogenesis. General Results: The CBA mouse strain has served as a valuable model for radiation-induced acute myeloid leukemia (AML) with respect to dose response relationships as well as underlying mechanisms. These mice have a very low spontaneous incidence of AML (<1%) but have been shown to be quite sensitive to induction of AML following exposure to gamma rays, x-rays and neutrons, with an RBE of 3-5 for fission spectrum neutrons. Another advantage is that few competing causes of death have been reported to occur in irradiated CBA mice. Their spontaneous frequency of other tumors is also relatively low except for liver tumors which have a frequency of approximately 12%. However, there have been no reports of an increase in liver tumor incidence following radiation exposure. This study was designed to compare the effects of Cs137 gamma rays with 1 GeV/n iron particles on the dose response for induction of AML in male CBA mice. Doses for

	gamma rays were based on previous studies and ranged from 1-3 Gy. For 1 GeV/n iron, doses selected were based on the assumption that the RBE for iron would be near that for neutrons, i.e., 3-5. On this basis doses ranging from 0.1 to 1 Gy were selected. As might be expected, the RBE for life shortening was estimated to be 3.7. However, unexpectedly, the RBE for AML induction by 1 GeV/n iron particles approximated 1. In addition, unlike previous studies and the present study which found very little effect of gamma-ray irradiation on the frequency of hepatocellular carcinomas, a large increase in the incidence of these tumors was found after iron irradiation where an approximate RBE of 35 was observed. These data suggest a difference in effects of HZE iron ions on the induction of leukemia compared to solid tumors suggesting potentially different mechanisms of tumorigenesis. The basis for these differences and the impact of these results on risk estimates for space travel are under investigation using cytogenetic and molecular endpoints including chromosome aberrations, expression profiling, comparative genomic hybridization (CGH), and microsatellite instability. These data are summarized below. Bone marrow cells from CBA are more sensitive than C57BL for early PU.1 loss for both HZE Fe and g-ray exposures. PU.1 loss reasonably follows expectations from gross aberration induction. At later times after either radiation, levels of PU.1 loss return to background levels in C57BL but not CBA mice. PU.1 loss at 1 month is suggested as a biomarker for sensitivity to AML induction. Radiation-induced chromosomal instability, using delayed chromatid-type aberrations as a biomarker in a spleen colony assay, appears higher in CBA than C57BL mice. Increased levels of chromatid aberrations were seen in AMLs arising from both radiations. No instabilities resulting in chromosomal translocations were seen in spleen colonies. Evidence of microsatellite instabilities was seen in AMLs and in spleen colonies for both radiation types. Development of a system for studying changes in human bone marrow cells in humanized NOD/SCID mice appears feasible. Analysis with CGH have identified two regions with consistent changes in murine AML and alterations in gene expression in these regions confirms this observation. Studies are also directed at human AML with sample from individuals with radiation-induced AML following therapy and using a humanized mouse model. Both of which have shown promising results. The key findings of this project to date are: 1) In mice, 1 GeV/n 56Fe ions do not appear to be substantially more effective than gamma-rays; 2) In contrast, 56Fe irradiated mice had much higher risk for hepatocellular carcinoma than gamma-ray irradiated mice; 3) Chromosome 2 deletions detected in bone marrow cells one month post-irradiation may be a biomarker for AML risk; 4)There are no apparent differences between Fe and gamma-ray induced AML at the cytogenetic or molecular level.
Bibliography Type:	Description: (Last Updated: 06/10/2025)
Abstracts for Journals and Proceedings	Peng Y, Warner CL, Liu X, Benik PC, Callan MA, Ray FA, Weil MM, Ullrich RL, Bedford JS. "Leukemogenesis and Early Loss of PU.1 on Chromosome 2 in CBA/CaJ and C57BL/6 Mice after Irradiation with HZE Iron Ions." 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, California, July 13-15, 2007. 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, California, July 13-15, 2007. p 11. , Jul-2007
Abstracts for Journals and Proceedings	Ullrich R, Bedford JS, Peng Y, Ray FA, Bacher JW, Nagarajan L, Story MD, Belmont JW, Cai WW, Erhart EJ, Genik PC, Bailey SM, Weil MM. "Radiation Leukemogenesis NSCOR." 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, California, July 13-15, 2007. 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, California, July 13-15, 2007. , Jul-2007
Abstracts for Journals and Proceedings	Peng Y, Warner C, Liu X, Genik P, Callan M, Ray FA, Weil M, Ullrich R, Bedford J. "Leukemogenesis and early loss of PU.1 in CBA mice after irradiation with gamma rays or HZE Fe ions." Thirteenth International Congress of Radiation Research, San Francisco, California, July 8-12 2007. Thirteenth International Congress of Radiation Research, San Francisco, California, July 8-12 2007. , Jul-2007
Abstracts for Journals and Proceedings	Maragnon D, Bailey SM, Muhlmann MC, Bedford JS. "Investigation of hot-spots associated with elements in the breakpoint cluster region surrounding spfi.1 (PU.1) gene deletions on chromosome 2 in radiation induced AML in CBA mice." Thirteenth International Congress of Radiation Research, San Francisco, California, July 8-12 2007. Thirteenth International Congress of Radiation Research, San Francisco, California, July 8-12 2007. , Jul-2007
Abstracts for Journals and Proceedings	Xiao G, Nagasawa H, Bouffler SD, Degg N, Peng Y, Ray FA, Roby A, Ullrich R, Bedford J, Weil M. "Characterization of radiobiological endpoints in cells from RI mice. Leukemogenesis and early loss of PU.1 in CBA mice after irradiation with gamma rays or HZE Fe ions." Thirteenth International Congress of Radiation Research, San Francisco, California, July 8-12 2007. Thirteenth International Congress of Radiation Research, San Francisco, California, July 8-12 2007. , Jul-2007
Abstracts for Journals and Proceedings	Ullrich R, Bedford JS, Peng Y, Ray FA, Bacher JW, Nagarajan L, Story MD, Belmont JW, Cai WW, Ehrhart EJ, Genik PC, Bailey SM, Weil MM. "Space Radiation Carcinogenesis: Reducing the Uncertainties." Invited Presentation. American Society for Gravitational and Space Biology Annual Meeting, Ames Research Center, October, 2007. Gravitational and Space Biology 2007 Oct;21(1):40. , Oct-2007